Polar Coordinates: Cartesian Equations – Part 2

Questions 1–11

Question 1. Write the following polar equations in Cartesian form, giving your answers in the form $ax + by = k$ where a, b and k are integers or surds to be found:

a) $r = \sec\left(\theta - \frac{\pi}{3}\right)$
b) $r = \sec\left(\theta + \frac{\pi}{6}\right)$
c) $r = \operatorname{cosec}\left(\theta + \frac{\pi}{4}\right)$
d) $r = 4\operatorname{cosec}\left(\theta - \frac{\pi}{6}\right)$

Question 2. Find the Cartesian equation for each of the following curves:

a) $r = 3 + \cos 2\theta$
b) $r^2 = \sin 2\theta$
c) $r^2 = 3\sec 2\theta$
d) $r^2 = 9\operatorname{cosec} 2\theta$
e) $r = 6\cot\theta\operatorname{cosec}\theta$

Question 3. The Cartesian equation

$$(x^2 + y^2)^2 = x^2 - y^2$$

when written in Polar form is given by $r^2 = \mathrm{f}(\theta)$. Find $\mathrm{f}(\theta)$ in its simplest form.

Question 4. The Cartesian equation $y = \frac{1}{x}$ when written in Polar form is given by $r^2 = \mathrm{f}(\theta)$. Find $\mathrm{f}(\theta)$ in its simplest form.

Question 5. It is known that the complex number z satisfies

$$|z - 6 + 8i| = 10$$

a) Sketch the locus of z on an Argand diagram.
b) Show that this locus of points can be represented by the polar curve

$$r = 12\cos\theta - 16\sin\theta$$

Question 6.
a) Show on an Argand diagram the locus of points by the values of z satisfying

$$|z - 4 - 3i| = 5$$

b) Show that this locus of points can be represented by the polar curve

$$r = 8\cos\theta + 6\sin\theta$$

Question 7. It is known that the complex number z satisfies

$$|z-1| = |z-\sqrt{3}i|$$

a) Sketch the locus of z on an Argand diagram.
b) Show that this locus of points can be represented by the polar curve

$$r = \frac{1}{2}\operatorname{cosec}\left(\theta - \frac{\pi}{6}\right)$$

Question 8.

a) Sketch the polar curve whose polar equation is $r = 1 + 2\cos 2\theta$ for $0 \leq \theta \leq 2\pi$, demonstrating any negative parts using a dashed line.
b) Prove that the values of θ at the points of intersection of the curve $r = 1 + 2\cos 2\theta$ with the line $r = \sqrt{3}\sec\theta$ are satisfied by

$$4\cos^3\theta - \cos\theta - \sqrt{3} = 0$$

Prerequisite Knowledge for Q9: Hyperbolic Functions and Hyperbolic Identities

Question 9. A curve C is given by

$$r = \sqrt{1 + \sinh^2 \theta}$$

A line L, given by $y = x$, intersects the curve C
Find the first two points of intersection between the line and the curve, giving your answers in exact form.

Question 10. A curve C has Cartesian equation

$$x^3 - y^3 = 3xy \text{ where } x \leq 0 \text{ and } y \geq 0$$

The equation of the curve C in Polar form is given by $r = \mathrm{f}(\theta)$ where $\frac{\pi}{2} \leq \theta \leq \pi$

a) Find $\mathrm{f}(\theta)$ giving your answers in terms of $\sin\theta$ and $\cos\theta$
b) Find an expression for $\mathrm{f}\left(\frac{3\pi}{2} - \theta\right)$
c) Hence find the line of symmetry of C in its Cartesian form.
d) Show that C has a loop.

Question 11. A curve C is given by

$$13x^2 - 6\sqrt{3}xy + 7y^2 = 1$$

a) Show that

$$13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta = 10 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta$$

b) Without using Calculus, find the greatest and least distances of points on the curve from the origin.